

ASR Shootout — Bangalore Locality Benchmark

Evaluation of Deepgram Nova-2, Whisper large-v3, and Whisper Hindi FT for Indian Conversational Speech

1. Approach

Model Selection

Three ASR systems were benchmarked:

Model	Type	Rationale
Deepgram Nova-2	API (baseline)	Required baseline; production-grade, low latency, Hindi support
Whisper large-v3	Open-source (OpenAI)	Best-in-class multilingual model; standard benchmark reference
Whisper Hindi FT (<code>vasista22/whisper-hindi-large-v2</code>)	Open-source (fine-tuned)	Whisper large-v2 fine-tuned on Hindi data; tests whether India-specific fine-tuning helps

The choice of Whisper Hindi FT over a generic third API was deliberate — a direct comparison between a multilingual model (Whisper large-v3) and its Hindi fine-tuned variant isolates the effect of domain adaptation, which is the core question for an Indian hiring platform.

Dataset

20 audio samples recorded on a phone mic in natural conversational Hinglish sentences (e.g. *"Haan bhai, main Koramangala mein rehta hoon"*), covering 20 Bangalore locality names. Recordings were varied across four conditions:

- **Quiet** (samples 1–7): controlled indoor environment
- **Noisy** (samples 8–13): background noise, traffic
- **Rushed** (samples 14–17): fast, hurried speech
- **Whispered** (samples 18–20): low-volume speech

All models output Hindi in Devanagari script. Ground truth was maintained in Devanagari for fair WER comparison.

Metrics

- **WER (Word Error Rate):** Standard transcription accuracy. Lower is better.

- **Entity Hit Rate (Strict):** Exact string match — did the locality name appear verbatim in the transcript.
- **Entity Hit Rate (Fuzzy):** Partial ratio match using RapidFuzz at threshold 80/100 — catches phonetic near-misses like मारता अली → मराठाहल्ली. This is the production-relevant metric.
- **Latency:** Wall-clock time per inference (API round-trip for Deepgram; local GPU for Whisper on Colab T4).

WER alone is insufficient here. A model scoring 0.3 WER but missing the locality name is less useful than one scoring 0.5 WER that captures it correctly. Strict entity matching (5% across all models) massively underestimates real-world utility — fuzzy matching better reflects what a post-processing pipeline can recover.

2. Key Findings

Overall Results

Model	Mean WER ↓	Entity Hit Rate (Strict)	Entity Hit Rate (Fuzzy) ↑	Avg Latency
Deepgram Nova-2	0.538	5.0%	40.0%	0.72s
Whisper large-v3	0.354	0.0%	40.0%	1.21s
Whisper Hindi FT	0.265	5.0%	50.0%	2.95s

Whisper Hindi FT wins on both transcription accuracy and entity extraction with 51% lower WER and 10 percentage points higher fuzzy entity hit rate than Deepgram. However it is 4x slower — a real production constraint for real-time telephony.

The gap between strict (5%) and fuzzy (40–50%) entity hit rates is itself a finding: **all three models are producing phonetically close but orthographically wrong locality names**. This means a fuzzy post-processing layer over any of these models is not optional — it is required for the system to be usable.

Results by Condition (Fuzzy Entity Hit Rate)

Condition	Deepgram	Whisper large-v3	Whisper Hindi FT
Quiet	71.4%	28.6%	71.4%
Noisy	16.7%	33.3%	33.3%
Rushed	0.0%	50.0%	25.0%

Condition	Deepgram	Whisper large-v3	Whisper Hindi FT
Whispered	66.7%	—	66.7%

Condition WER Breakdown:

Condition	Deepgram WER	Whisper v3 WER	Hindi FT WER
Quiet	0.418	0.315	0.183
Noisy	0.619	0.345	0.314
Rushed	0.675	0.487	0.356
Whispered	0.476	0.286	0.238

Whisper Hindi FT has the best WER in every single condition without exception.

3. Failure Analysis

Where Models Break

Phonetic corruption of locality names is the dominant failure mode across all models:

Locality	Deepgram	Whisper large-v3	Whisper Hindi FT
मराठाहल्ली	मार ताली	मारताली	मारता अली
हेब्बल	है बल	फैबल	ठेब्बल
राजाजीनगर	रजनी नगर	राजिने नागर	राजनीतिक
व्हाइटफील्ड	white field	वाइट फिल्म	व्हाइट फील्ड
बोम्मनहल्ली	बोमन हेली	बम्मन हली	वोमनहली

Three distinct failure patterns:

1. **Phonetic near-misses (most common):** Model hears something close but orthographically wrong — मारता अली for मराठाहल्ली. Recoverable with fuzzy matching.
2. **Semantic hallucination (Whisper models):** Whisper large-v3 transcribes राजाजीनगर as राजिने नागर and व्हाइटफील्ड as वाइट फिल्म — substituting phonetically similar but meaningless words. Known Whisper failure mode on rare proper nouns.

3. Complete dropout under noise: Sample 11 (एचएसआर लेआउट, noisy) — Deepgram returned an empty string, Whisper Hindi FT returned only `वहाँ हूँ मैं`. The locality was entirely dropped from the transcript.

Surprising Findings

Rushed speech is the hardest condition, not noise:

- Deepgram scores 0% entity hit rate on rushed speech — complete failure
- This matters because candidates calling from construction sites or while commuting will often speak fast
- Whisper large-v3 surprisingly handles rushed speech better (50%) than quiet (28.6%)

Whispered speech is handled better than expected:

- Deepgram and Whisper Hindi FT both achieve 66.7% on whispered samples
- Better than noisy conditions for both models
- Suggests these models have seen whispered speech in training data

Deepgram's mixed-script output is a silent killer:

- Deepgram outputs inconsistent scripts — `"office white field में है"`, `"btm लेआउट"`
 - This makes downstream entity matching harder even when transcription is partially correct
 - A production system using Deepgram needs script normalisation before entity extraction
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4. Recommendation

No model is production-ready as-is for raw locality extraction. Even with fuzzy matching, the best result is 50% — meaning half of candidate locations are still being missed.

Immediate (ship now)

Use **Deepgram Nova-2** as the transcription layer — 0.72s latency is the only viable option for real-time phone calls — paired with a **fuzzy string matching post-processor** against a curated list of Bangalore localities. Fuzzy matching alone lifts entity hit rate from 5% to 40%. Deepgram also supports streaming natively; Whisper does not.

Medium-term (if accuracy matters more than latency)

Fine-tune Whisper Hindi FT on a dataset of Bangalore locality names spoken in conversational Hinglish. The base model already has the best WER (0.265) and highest fuzzy entity hit rate (50%). Domain-specific fine-tuning on even 200–300 locality utterances would likely push entity hit rate above 80%.

Do not use Whisper large-v3 for this task

Zero strict entity hits across all conditions. Its multilingual generality actively hurts performance on Indian proper nouns. Whisper Hindi FT strictly dominates it on every metric.

Cost at Scale

Model	Cost	Infra	Streaming
Deepgram Nova-2	~\$0.0043/min	Fully managed	 Yes
Whisper large-v3	Compute cost only	Self-hosted GPU	 No
Whisper Hindi FT	Compute cost only	Self-hosted GPU	 No

For a platform handling thousands of daily calls, Deepgram's managed infrastructure and streaming support justify its higher cost over self-hosted Whisper — until a fine-tuned model is ready to deploy.

Limitations

- 20 samples from a single speaker is a small dataset. Results may not generalise across Telugu-accented or Kannada-accented Hindi.
- Latency for Whisper models measured on Colab T4 GPU — production numbers will vary significantly by hardware.
- Fuzzy matching threshold (80/100) was set empirically; production tuning on a larger dataset is needed.
- No open-source dataset testing (Kathbath, Mozilla Common Voice) was performed — results reflect only self-recorded samples.